

Problem

Design a parallel RLC circuit that has the characteristic equation

$$s^2 + 100s + 10^6 = 0.$$

Step-by-step solution

Step 1 of 4

The parallel RLC circuit is shown in Figure 1.

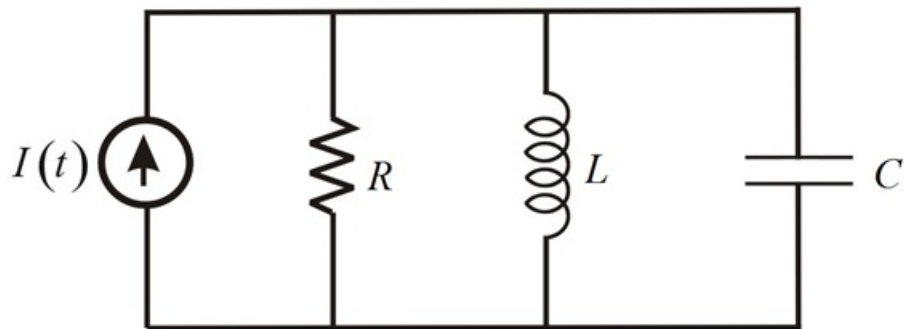


Figure 1

The parallel RLC circuit has the characteristic equation.

$$s^2 + 100s + 10^6 = 0 \dots\dots (1)$$

Step 2 of 4

Frequency domain circuit of Figure 1 is shown in Figure 2.

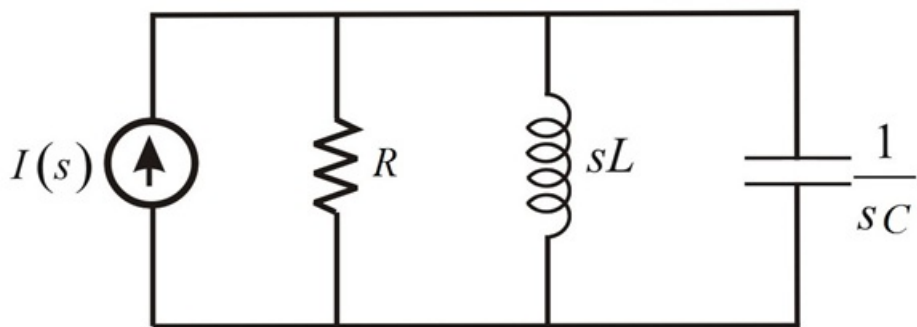


Figure 2

Step 3 of 4

The admittance Y of Figure 2 is,

$$Y = \frac{1}{R} + \frac{1}{sL} + sC$$

$$= \frac{sL + R + s^2 LCR}{RLs}$$

The impedance Z is the reciprocal of the admittance Y .

$$Z = \frac{1}{Y}$$

$$Z = \frac{RLs}{s^2 RLC + Ls + R}$$

$$Z = \frac{\frac{1}{C}s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}} \dots\dots (2)$$

The characteristic equation is the denominator polynomial of equation (1) equals to zero.

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \dots\dots (3)$$

Step 4 of 4

Compare equations (1) and (3).

$$\frac{1}{LC} = 10^6 \dots\dots (4)$$

$$\frac{1}{RC} = 100 \dots\dots (5)$$

Assume 10 mF for C . Substitute the value of C in equations (4) and (5).

$$\frac{1}{L(10)(10^{-3})} = 10^6$$

$$L = \frac{1}{(10^6)(10^{-2})}$$

$$= 0.1 \text{ mH}$$

The value of inductance is 0.1 mH.

$$\frac{1}{R(10)(10^{-3})} = 100$$

$$R = \frac{1}{(10)(10^{-3})(100)}$$

$$= 1 \Omega$$

The value of resistance is 1 Ω .

Therefore, the parallel circuit is shown in Figure 3.

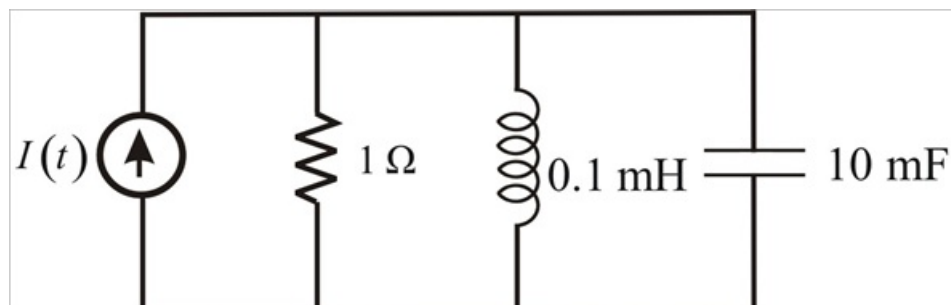


Figure 3